Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Andaya, JamiHyari D.
BSIT 4A

BIOS vs UEFI Comparison

Comparison Table

BIOS vs UEFI Comparison — Paraphrased Table

Comparison of traditional BIOS and modern UEFI firmware

Category	BIOS	UEFI
Definition	An older firmware system that initializes computer hardware and starts the operating system boot process.	A newer firmware interface that connects the operating system with the computer's firmware.
Boot Process	Runs a Power-On Self-Test, detects storage devices, reads boot information from the MBR, and transfers control to the bootloader.	Initializes hardware and uses its built-in boot manager to locate and start the operating system from the EFI System Partition.
Maximum Disk Support	When used with MBR, boot drives are generally limited to about 2 TB with standard 512-byte sectors.	Usually works with GPT, allowing support for drives larger than 2 TB and providing a much greater theoretical capacity.
Partition Style	Traditionally uses the Master Boot Record (MBR), which normally supports up to four primary partitions.	Commonly uses the GUID Partition Table (GPT), which supports larger drives and more flexible partitioning.
Security Features	Provides basic startup protection but normally cannot verify whether boot software is trusted.	Supports Secure Boot, which checks digital signatures to help prevent unauthorized or malicious software from loading during startup.
Speed	May take longer to initialize hardware and locate boot devices because it uses an older boot method.	Can provide faster startup through more efficient hardware initialization and direct access to boot files.
Advantages	Simple, familiar, and compatible with many older operating systems and hardware devices.	Supports larger drives, more partitions, Secure Boot, graphical interfaces, and advanced firmware features.
Disadvantages	Has limitations in disk capacity, partitioning, security, boot speed, and expandability.	Offers more features but may be more complicated to configure and can have compatibility issues with older systems.
Modern Usage	Mostly found in legacy computers and systems that need compatibility with older hardware or software.	The current firmware standard used by mos modern desktops, laptops, workstations, and servers.

The Shift from BIOS to UEFI

For many years, the Basic Input/Output System (BIOS) served as the standard firmware interface for computers. It was responsible for checking hardware during startup, identifying boot devices, and transferring control to the operating system. BIOS worked effectively with older generations of hardware, but its limitations became more noticeable as computers and storage technologies developed.

UEFI was introduced as a more modern alternative designed to address many of these limitations. It provides improved hardware initialization, better support for modern storage devices, enhanced security features, and a more flexible boot process. Because of these improvements, UEFI has become the standard firmware interface on most modern computer systems.

Key Advantages of UEFI over Traditional BIOS

Greater Storage Capacity: BIOS commonly uses the Master Boot Record (MBR), which limits supported boot disks to around 2 TB and allows only a limited number of primary partitions. UEFI commonly uses the GUID Partition Table (GPT), which supports much larger drives and more flexible partitioning.

Improved Boot Management: Instead of depending on traditional boot code stored in the MBR, UEFI includes its own firmware-based boot manager. It can access boot entries stored on the EFI System Partition, making it easier to manage multiple operating systems and boot configurations.

Stronger Startup Security: Traditional BIOS has limited methods for checking whether startup software is trustworthy. UEFI provides Secure Boot, which verifies digitally signed boot components before allowing them to run. This helps protect the system against unauthorized startup software.

More Flexible Firmware Features: Unlike the basic text-based BIOS interface, UEFI can provide graphical configuration screens, mouse support, network boot capabilities, modular drivers, and additional firmware management features.

Drawbacks and Present-Day Relevance

Although UEFI is now widely used, it also has some limitations. Its larger number of features can make configuration more complicated, especially for beginners. Secure Boot can also cause compatibility problems when using older operating systems or unsigned third-party drivers.

Summary

UEFI has largely replaced traditional BIOS because it provides better support for modern storage devices, faster startup, improved security, and more advanced firmware capabilities. However, BIOS remains relevant for older or legacy systems. For system administrators, understanding both BIOS and UEFI is important because they may need to manage a mixture of older and newer computer systems.

References

Microsoft. (2026). Windows support for hard disks that are larger than 2 TB. Microsoft Learn.

Microsoft. (2026). UEFI/GPT-based hard drive partitions. Microsoft Learn.

Ubuntu. (n.d.). UEFI Secure Boot. Ubuntu Security Documentation.

Unified EFI Forum. (2024). Unified Extensible Firmware Interface Specification, Version 2.10.